

Industrial Internship Report on "PREDICTIVE MAINTENANCE SYSTEM"

Prepared by

VISHAL R. KOTALWAR

Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time.

My project was : IoT-Based Predictive Maintenance System using ESP32

This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

TABLE OF CONTENTS

1	Preface	3
2	Introduction	4
2.1	About UniConverge Technologies Pvt Ltd	4
2.2	About upskill Campus	8
2.3	Objective	10
2.4	Reference	10
2.5	Glossary	10
3	Problem Statement	12
4	Existing and Proposed solution	13
5	Proposed Design/ Model	14
5.1	High Level Diagram (if applicable)	14
5.2	Low Level Diagram (if applicable)	14
5.3	Interfaces (if applicable)	14
6	Performance Test	16
6.1	Test Plan/ Test Cases	16
6.2	Test Procedure	17
6.3	Performance Outcome	17
7	My learnings	18
8	Future work scope	18

1 Preface

This internship provided an excellent opportunity to gain practical exposure to real-world industrial problems and their solutions. The program was organized by upskill Campus in collaboration with UniConverge Technologies Pvt Ltd (UCT) and The IoT Academy. The internship duration was 6 weeks, during which I worked on a project focused on predictive maintenance using IoT.

The project aimed to design and implement a system capable of monitoring machine health parameters and providing early warnings for potential failures. This helped in understanding the importance of preventive strategies in industrial environments.

Throughout this internship, I learned about IoT systems, embedded programming, and real-time data monitoring. I also gained experience in problem-solving, system design, and technical documentation.

I would like to express my gratitude to all the mentors and instructors who guided me throughout this internship. Their support and insights were invaluable. This experience has significantly contributed to my technical growth and career development.

.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoraWAN), Java Full Stack, Python, Front end** etc.



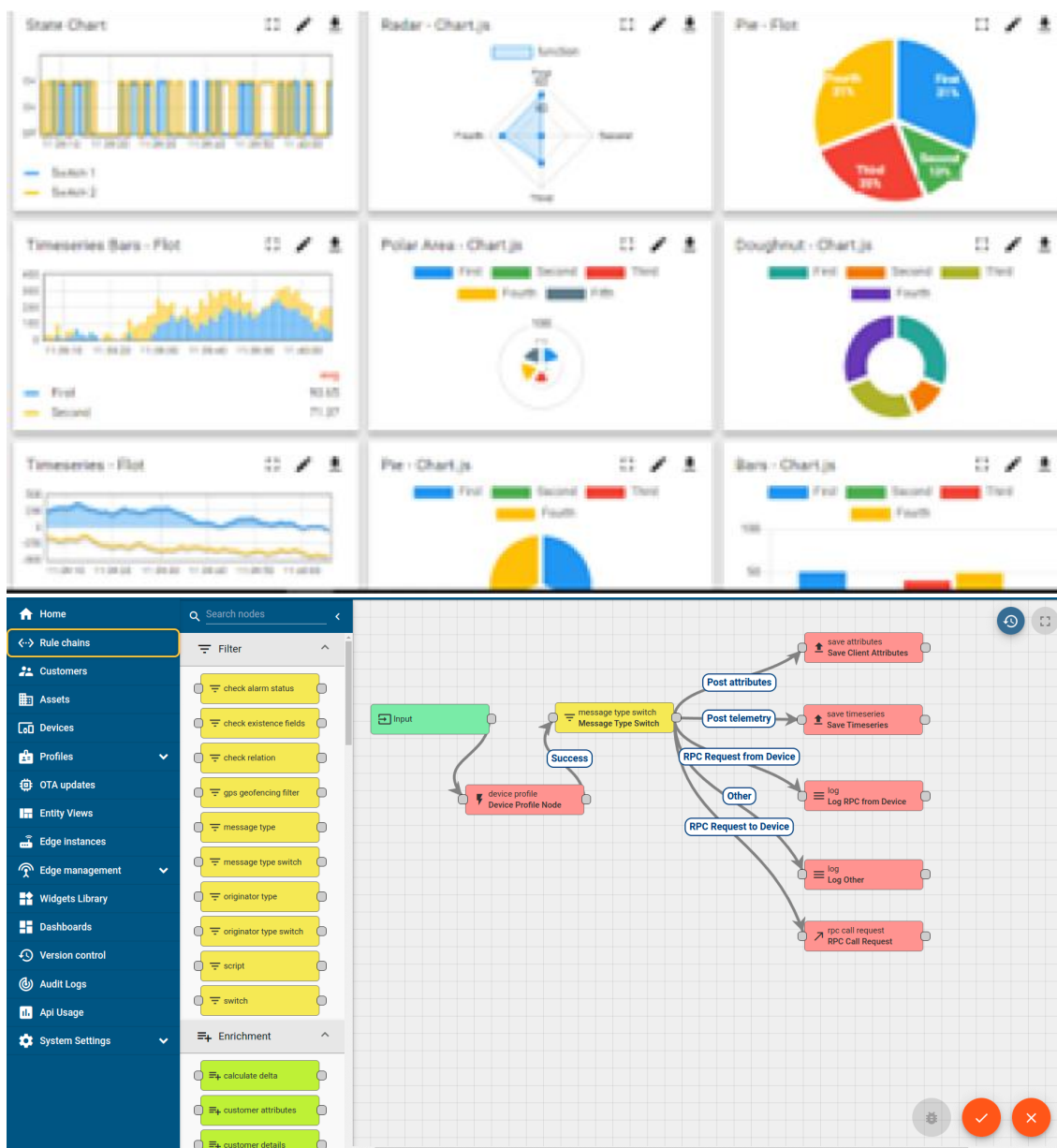
i. UCT IoT Platform ()

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



ii. **Smart Factory Platform (**FACTORY WATCH**)**

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Prod	Downtime	Idle		
CNC_S7_81	Operator 1	WO040520001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO040520001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i



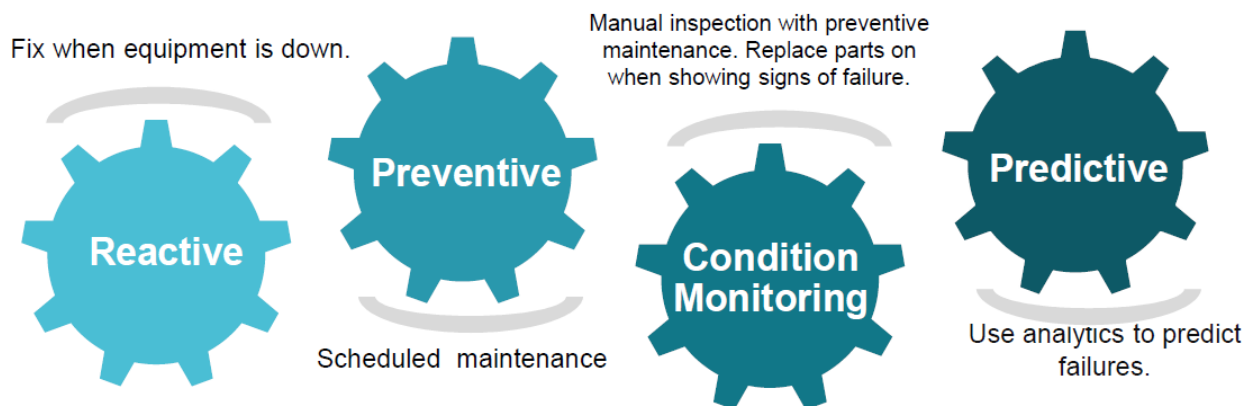


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

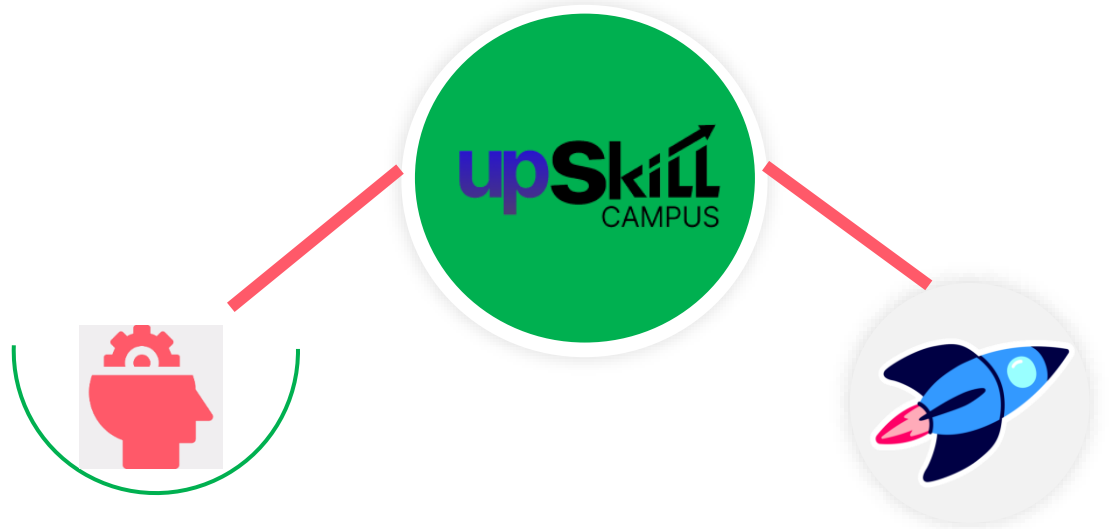
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>

Career growth/upskilling

- Interview Preparation and skill building
- upskilling Courses
- Skill Assessment
- Profile building

Professional networking

- Alumni Connections
- Mentorship
- Discussion/QA forum

Collaboration platform

- Project collaboration
- Discussion forum
- Tech updates

Job/internship platform

- Job portal
- Internship portal
- Freelancing projects

2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

2.5 Reference

- [1] IoT documentation and tutorials
- [2] ESP32 programming guides
- [3] Online resources and research papers

2.6 Glossary

Terms	Acronym
IoT	Internet of Things
ESP32	Microcontroller with WiFi
Sensor	Device to measure physical parameters
Cloud	Online data storage and processing

Predictive Maintenance	Failure prediction before occurrence
---------------------------	--------------------------------------

3 Problem Statement

Industries often face unexpected machine failures that lead to production downtime, increased maintenance costs, and reduced efficiency. Traditional maintenance approaches such as reactive and scheduled maintenance are not efficient as they either respond after failure or do not consider real-time machine conditions.

The objective of this project is to design an IoT-based system that continuously monitors machine parameters such as temperature and vibration and provides early warnings when abnormal conditions are detected

4 Existing and Proposed solution

4.1 Existing Solutions:

- Reactive Maintenance (after failure)
- Preventive Maintenance (scheduled checks)

4.1.1 Limitations:

- High downtime
- Increased cost
- No real-time monitoring

4.2 Proposed Solution:

The proposed system uses IoT technology to monitor machine conditions in real-time using sensors connected to ESP32. It analyzes data using predefined thresholds and generates alerts when abnormal values are detected.

4.2.1 Value Addition:

- Real-time monitoring
- Low-cost solution
- Early fault detection
- Easy scalability

4.3 Code submission <https://github.com/vishal16705/upskillcampus.git>

5 Proposed Design/ Model

5.1 High Level Diagram



Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram

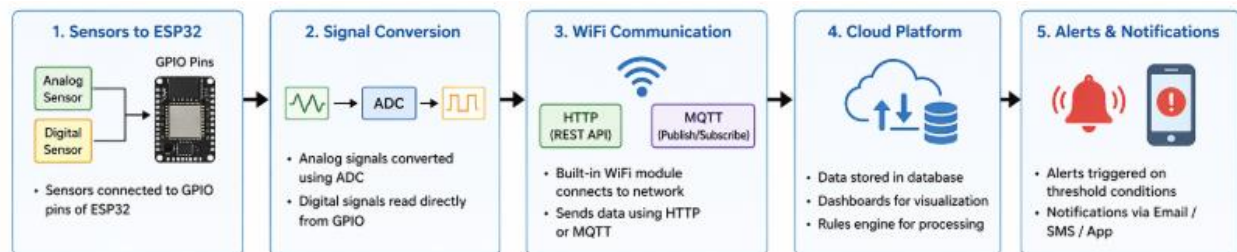


Figure 2: LOW LEVEL DIAGRAM OF THE SYSTEM

5.3 Interfaces

1. Sensor Interface



- Supports analog and digital inputs
- Connected via GPIO pins
- Includes signal conditioning if required

2. WiFi Communication Interface



- Enables wireless data transmission
- Supports:
 - HTTP (REST-based communication)
 - MQTT (publish/subscribe model)

3. Cloud API Interface



- Receives data from ESP32
- Provides endpoints for:
 - Data storage
 - Data retrieval
 - Alert configuration
- Integrates with dashboards and notification systems

6 Performance Test

The performance testing phase evaluates the effectiveness of the developed IoT-based predictive maintenance system in real-world conditions. This section focuses on identifying system constraints such as processing capability, sensor accuracy, communication latency, and power consumption.

The system is built using the **ESP32**, which provides sufficient computational power and wireless connectivity for real-time monitoring. However, certain constraints must be considered to ensure reliable performance in industrial environments.

6.1.1 Identified Constraints:

- **Memory Limitation:** Limited RAM and storage in embedded systems
- **Processing Speed (MIPS):** Real-time data processing capability
- **Sensor Accuracy:** Variations in sensor readings
- **Network Latency:** Delay in data transmission over WiFi
- **Power Consumption:** Continuous operation requirement
- **Scalability:** Handling multiple sensors/devices

6.1.2 Handling of Constraints:

- Optimized code to reduce memory usage
- Lightweight data processing on ESP32
- Use of calibrated sensors for better accuracy
- Efficient WiFi communication protocols
- Low-power operation modes of ESP32
- Modular design for scalability

6.2 Test Plan/ Test Cases

The following test cases were designed to evaluate system performance:

Test Case ID	Test Description	Expected Outcome
TC1	Sensor Data Acquisition	Accurate sensor readings

Test Case ID	Test Description	Expected Outcome
TC2	WiFi Connectivity	Stable connection to network
TC3	Data Transmission	Successful data upload to cloud
TC4	Threshold Detection	Alert triggered on abnormal values
TC5	System Response Time	Quick response (<2 seconds)
TC6	Continuous Operation	Stable performance over time

6.3 Test Procedure

The system was tested under controlled conditions to simulate real-world scenarios:

1. Sensors were connected to the ESP32 and calibrated
2. ESP32 was connected to a WiFi network
3. Data was continuously collected and transmitted to the cloud
4. Abnormal conditions were simulated:
 - Increasing temperature artificially
 - Introducing vibration disturbances
5. System behavior was observed for:
 - Data accuracy
 - Response time
 - Alert generation
6. Continuous testing was performed for extended duration to check stability

6.4 Performance Outcome

6.4.1 Observations:

- Sensor readings were consistent with acceptable accuracy
- WiFi communication remained stable during operation
- Alerts were generated successfully when thresholds were exceeded
- Average response time was within acceptable limits (~1–2 seconds)
- System operated continuously without failure

6.4.2 Limitations Observed:

- Minor fluctuations in sensor readings

- Dependency on stable internet connection
- No advanced prediction (only threshold-based detection)

7 My learnings

- Understanding of IoT architecture
- ESP32 programming and interfacing
- Real-time data monitoring
- Cloud integration basics
- Problem-solving and system design

8 Future work scope

- Integration of Machine Learning for better prediction
- Development of mobile application
- Advanced analytics and dashboards
- Implementation of predictive algorithms for remaining useful life

